

EXPERIMENT 4

Analysis of Continuous Charge Distribution in electric field. To find the (i) line charge density, (ii) Surface charge density and (iii) Volume charge density for any specific (a) line, (b) surface or (c) volume.

Aim: To write program for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB.

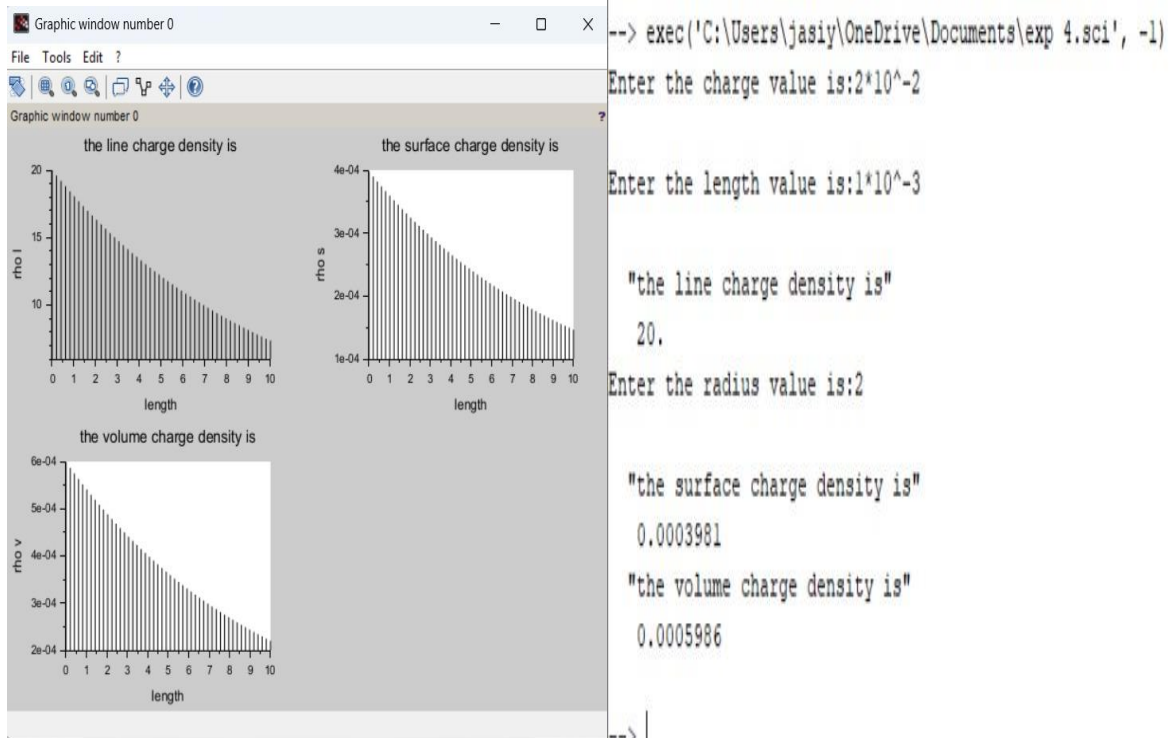
Software required:

- SCILAB version 6.0.1

Program:

```
EXP 1 EME.sci  Exp 1)E.sci  exp 1)eme.sci  exp 2.sci  exp 2 case 1.sci  exp 3.sci  exp 4.sci
1  clc;
2  clear;
3  q=input("Enter the charge value is:");
4  l=input("Enter the length value is:");
5  //line charge density
6  rl=(q/l);
7  disp('the line charge density is',[rl]);
8  //surface charge density
9  pi=3.14;
10 r=input("Enter the radius value is:");
11 rs=q/(4*pi*r^2);
12 disp('the surface charge density is',[rs]);
13 //volume charge density
14 rv=q/(1.33*pi*r^3);
15 disp('the volume charge density is',[rv]);
16 x = linspace(0, 10, 50);
17 y = rl*exp(-0.1 * x);
18 z = rs*exp(-0.1 * x);
19 v = rv*exp(-0.1 * x);
20
21 figure();
22 subplot(2, 2, 1);
23 plot2d3(x,y);
24 title('the line charge density is');
25 xlabel("length");
26 ylabel("rho.l");
27 subplot(2, 2, 2);
28 plot2d3(x,z);
29 title('the surface charge density is');
30 xlabel("length");
31 ylabel("rho.s");
32 subplot(2, 2, 3);
33 plot2d3(x,v);
34 title('the volume charge density is');
35 xlabel("length");
36 ylabel("rho.v");
37
38
```

Output:



Theoretical Calculation :

EXP NO: 04

Analysis of Continuous Charge Distribution in electric field.

To find (i) the charge density, (ii) Surface charge and (iii) Volume charge density for an specific coil line, (b) Surface or (c) volume.

Output :

Enter the charge value is : 2×10^{-2}

Enter the length value is : 1×10^{-3}

Enter the radius value is : 2

PRACTICAL OUTPUT:

"the line charge density is"

20.

"the surface charge density is"

0.0003981

"the volume charge density is"

0.0005986

THEORETICAL CALCULATION:

$$\rho_L = \frac{Q}{L} = \frac{2 \times 10^{-2}}{1 \times 10^{-3}} = \frac{0.02}{0.001} = 20 \text{ cm}.$$

Theoretical line charge density = 20 cm.

$$\rho_S = \frac{Q}{2\pi r L} = \frac{2 \times 10^{-2}}{2 \times 3.14 \times 1 \times 10^{-3} \times 2} = \frac{0.02}{0.012566}$$

$\rho_S = 1.5915 \text{ cm}^2$

Theoretical surface charge density $\rho_S = 1.5915 \text{ cm}^2$.

$$\rho_V = \frac{Q}{\pi r^2 L} = \frac{0.02}{3.14 \times 4 \times 0.001}$$

$\rho_V = 1.5915 \text{ cm}^3$

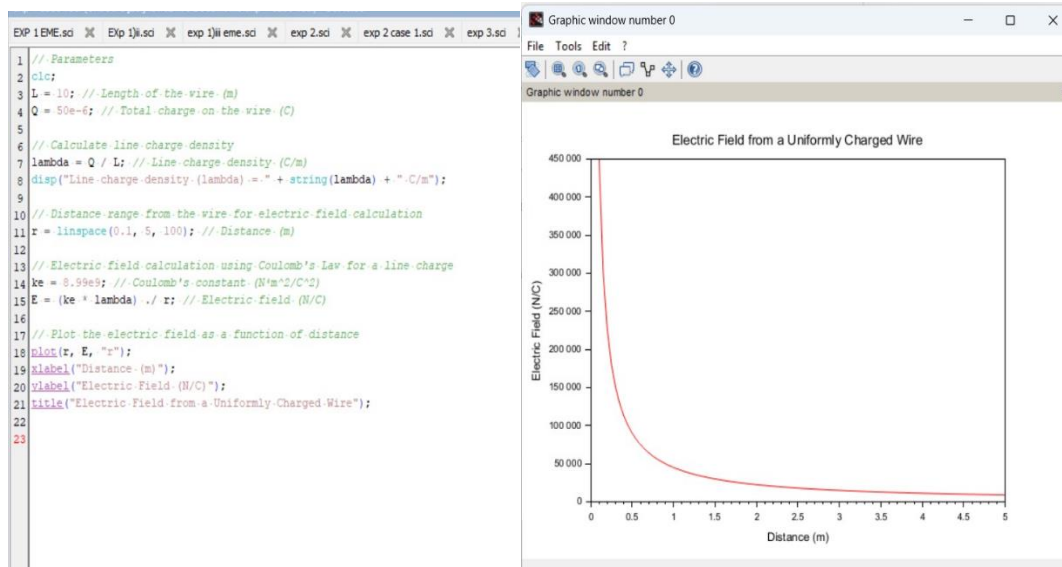
Theoretical volume charge density $\rho_V = 1.5915 \text{ cm}^3$

Case 1:

A long wire of length $L=10\text{m}$ carries a uniform charge of $Q=50\mu\text{C}$. The goal is to calculate the line charge density λ and determine how the electric field behaves as a function of distance from the wire.

Program:

Output:



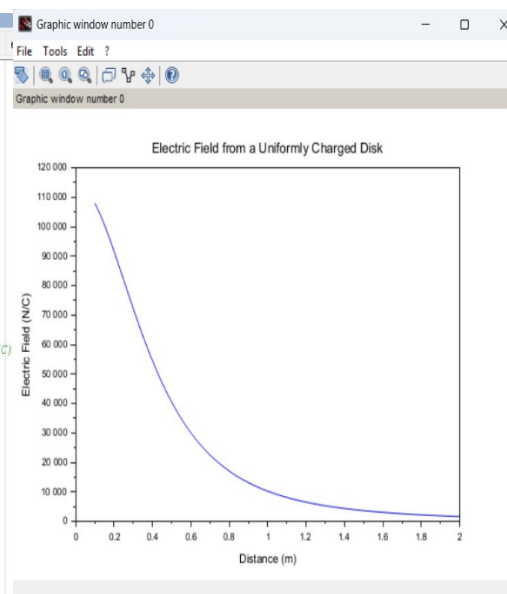
Case 2:

Given a uniformly charged disk of radius $R=0.5\text{m}$ with total charge $Q=10\mu\text{C}$, calculate the surface charge density and plot the electric field along the axis of the disk. Explain the behavior of the electric field as a function of distance from the center of the disk.

Code:

```
exp 4 case 1.sci (C:\Users\jasy\OneDrive\Documents\exp 4 case 1.sci) - SciNotes
EXP 1 EME.sci EXP 1 IJ.sci EXP 1 IJ EME.sci EXP 2.sci EXP 2 case 1.sci EXP 3.sci
1 // Parameters
2 clc;
3 R = 0.5; // Radius of the disk (m)
4 Q = 10e-6; // Total charge on the disk (C)
5
6 // Calculate surface charge density
7 A = pi * R^2; // Area of the disk (m^2)
8 sigma = Q / A; // Surface charge density (C/m^2)
9 disp("Surface charge density (sigma) = " + string(sigma) + " C/m^2");
10
11 // Distance from the center of the disk to calculate electric field
12 z = linspace(0.1, 2, 100); // Distance (m)
13
14 // Electric field on the axis of the disk
15 ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
16 E_disk = (ke * sigma * R^2) ./ (2 * (z.^2 + R^2).^(3/2)); // Electric field (N/C)
17
18 // Plot the electric field as a function of distance
19 plot(z, E_disk, "b");
20 xlabel("Distance (m)");
21 ylabel("Electric Field (N/C)");
22 title("Electric Field from a Uniformly Charged Disk");
23
```

Output:



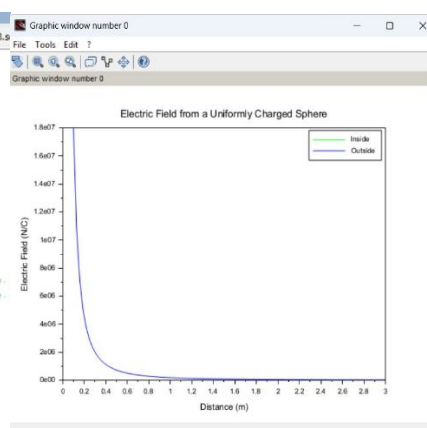
Case 3:

Given a uniformly charged sphere of radius $R=2\text{m}$ with total charge $Q=20\mu\text{C}$, calculate the volume charge density and plot the electric field both inside and outside the sphere. Explain the difference in electric field behavior for points inside and outside the sphere.

Code:

```
exp 4 case 1.sci (C:\Users\jasy\OneDrive\Documents\exp 4 case 1.sci) - SciNotes
EXP 1 EME.sci EXP 1 IJ.sci EXP 1 IJ EME.sci EXP 2.sci EXP 2 case 1.sci EXP 3.sci
1 // Parameters
2 clc;
3 R = 2; // Radius of the sphere (m)
4 Q = 20e-6; // Total charge (C)
5
6 // Calculate volume charge density
7 V = (4/3) * pi * R^3; // Volume of the sphere (m^3)
8 rho = Q / V; // Volume charge density (C/m^3)
9 disp("Volume charge density (rho) = " + string(rho) + " C/m^3");
10
11 // Distance from the center of the sphere
12 z = linspace(0.1, 3, 100); // Distance (m)
13
14 // Electric field inside and outside the sphere (using Gauss's Law)
15 ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
16 E_sphere_inside = (ke * rho * z) / 3; // Electric field inside the sphere
17 E_sphere_outside = (ke * Q) ./ z.^2; // Electric field outside the sphere
18
19 // Plot the electric field inside and outside the sphere
20 plot(z, E_sphere_inside, "g", z, E_sphere_outside, "b");
21 xlabel("Distance (m)");
22 ylabel("Electric Field (N/C)");
23 title("Electric Field from a Uniformly Charged Sphere");
24 legend("Inside", "Outside");
25
```

Output:



Result : Thus the program was verified for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB was successfully.

